

3.1.2 Energy generation and storage

Students should understand how energy is generated and stored and how this is used as the basis for the selection of products and power systems.

Fossil fuels

Content	Potential links to maths and science
How power is generated from: • coal • gas • oil. Arguments for and against the selection of fossil fuels.	How to choose appropriate energy sources.

Nuclear power

Content	Potential links to maths and science
How nuclear power is generated. Arguments for and against the selection of nuclear power.	How to choose appropriate energy sources.

Renewable energy

Content	Potential links to maths and science
How power is generated from: • wind • solar • tidal • hydro-electrical • biomass. Arguments for and against the selection of renewable energy.	How to choose appropriate energy sources.

Energy storage systems including batteries

Content	Potential links to maths and science
Kinetic pumped storage systems. Alkaline and re-chargeable batteries.	How to choose appropriate energy sources.